OR4023 (Optimization) Homework 2

Name: _____ Student ID: _____

Question 1. Let $f(x) = x^4 - 2x^3$.

(1) Find the global minimum of f over $[-1, 2]$ in the following steps.

   (a) Apply the first order necessary conditions for interior points to find candidates for local minimum points.

   (b) Apply the first order necessary conditions for the boundary points to find candidate for local minimum points.

   (c) Compare function values at the points in (a) and (b) to find the global minimum points.

(2) Apply the second order sufficient conditions to determine which points found in (1a) and (1b) are (strict) local minimum points.

Question 2. Find a strict local minimum of $f(x, y) = x^4 + 2y^2 - xy$ over $\mathbb{R}^2$ using the second order sufficient conditions.

Question 3. Let $b \in \mathbb{R}^n$ be a vector and $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix.

(a) If $f(x) = b^T x$, show $\nabla f(x) = b$ and $\nabla^2 f(x) = 0$.

(b) If $f(x) = \frac{1}{2}x^T Ax$, show $\nabla f(x) = Ax$ and $\nabla^2 f(x) = A$.

Question 4. Let

$$f(x_1, x_2) = x_1^2 + 4x_2^2 - 6x_1x_2 + 5x_1 - 7x_2 + 8.$$

Let $x = (x_1, x_2) \in \mathbb{R}^2$. Find a symmetric matrix A, a vector b and a number c so that

$$f(x) = \frac{1}{2}x^T Ax + b^T x + c.$$

Question 5. Let $f(x) = x^2 + e^x, x \in [-1, 1]$.

(a) Show that f has a unique minimizer, which is strict and global.

(b) Start from the initial point $x_0 = 0$, apply 2 iterations of the Newton's method to find x_2.

(c) Start from the initial points $x_0 = -1, x_1 = 1$, apply 2 iterations of the secant method to find x_3.

Question 6. For each of the following sequences, determine their limit and the order of convergence (or superlinear if there is no order).

(a) $r_k = \frac{1}{k} - \frac{1}{k+1}, k \geqslant 1$

(b) $r_{k+1} = \frac{r_k}{k+1}, k \geqslant 1$